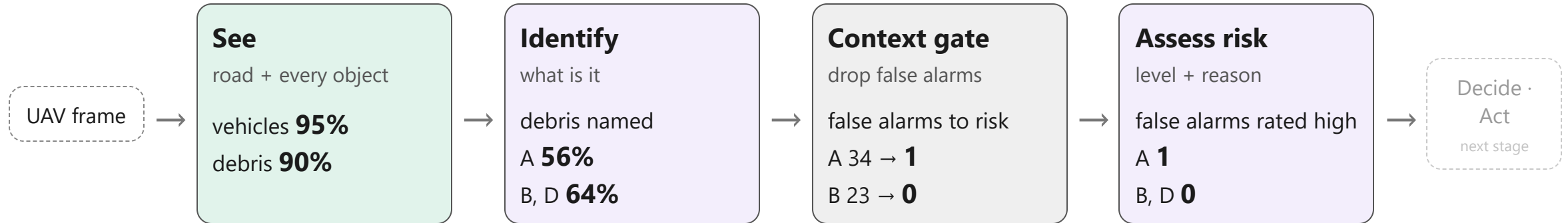


STAGE REPORT · 28 SEPTEMBER 2026

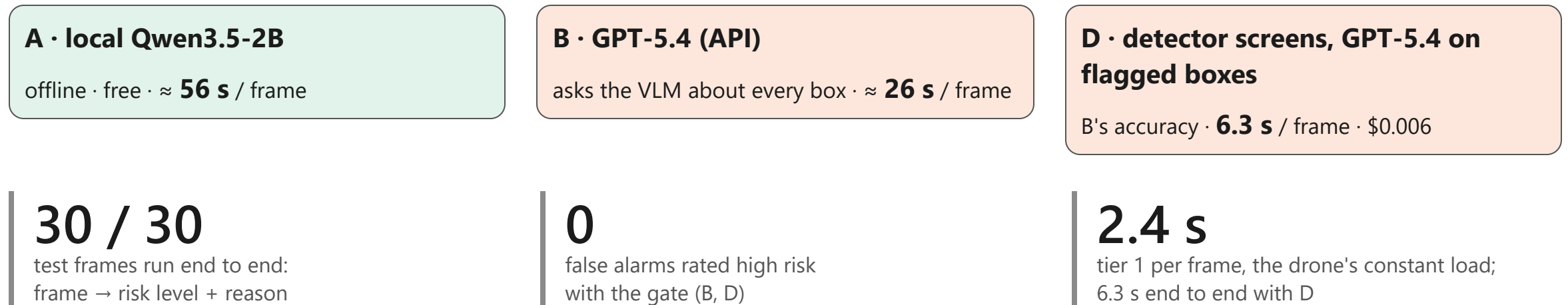
Highway hazard perception from a drone:
connecting the chain from seeing to risk, and comparing three architectures

Process visualization and architectures

① Chain connected end to end · all 30 test frames



② Three architectures compared



Real UAV motorway frames + rendered 3D debris



Background

real UAV video, Pexels clip 8742752
Pexels License · open motorway only

Debris (red)

3D model "Low-Poly Treaded Tire"
by TepidGames · CC BY · Sketchfab
Blender 5.2.2 · random pose
real-world scale from calibrated
ground resolution (px per m)

Debris label

pixel-exact, from the render's alpha

Vehicle labels (blue)

Grounding DINO proposals
every box checked by hand

4 open-motorway locations

30 test frames · fixed

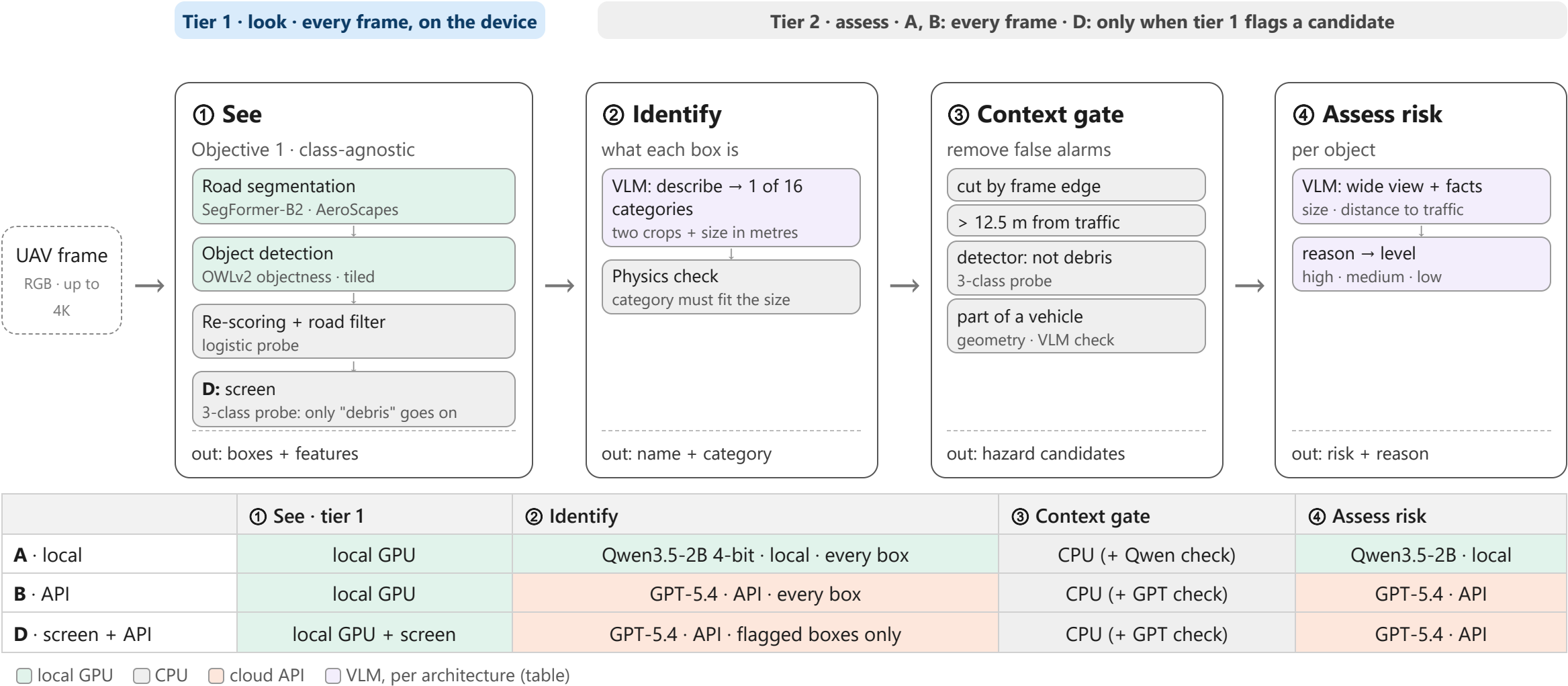
30 dev frames · all tuning

342 vehicles · hand-checked

1 debris object per test frame

leave-one-location-out scoring

Architectures

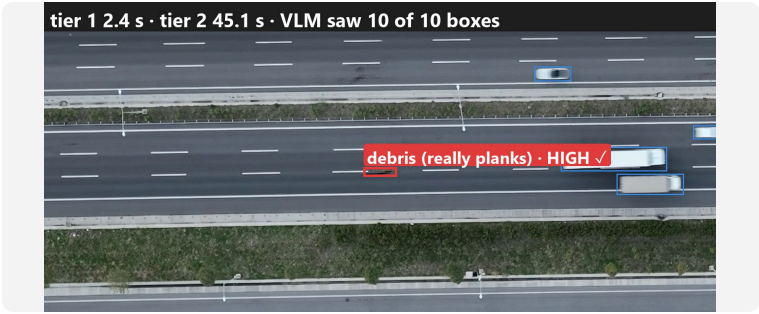


A · local Qwen3.5-2B

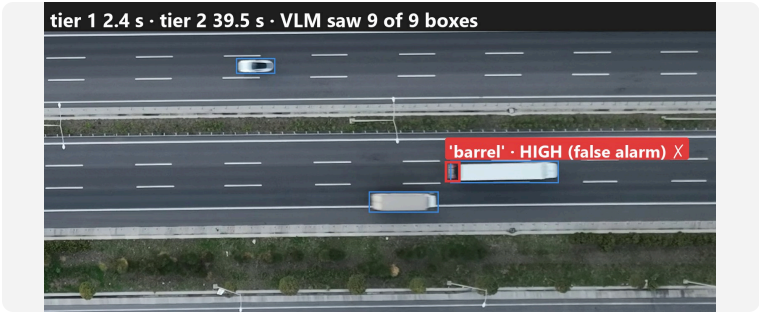
offline and free · alerts on the debris about as often as B · but misnames, over-rates and one false alarm · tier 2 12-129 s
■ high ■ medium ■ low ■ removed by gate ■ missed ■ vehicle · banner: this frame's tier 1 and tier 2 time, and boxes the VLM saw



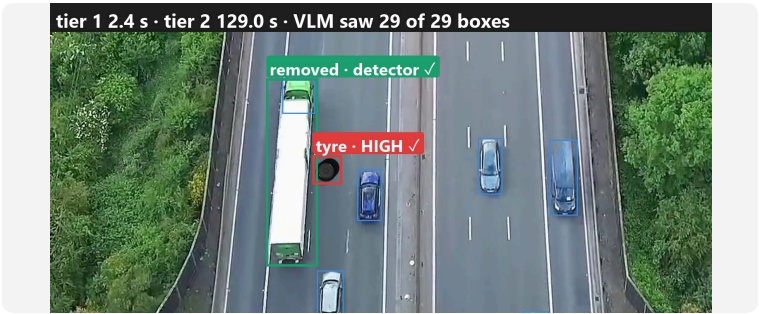
✓ tyre · HIGH



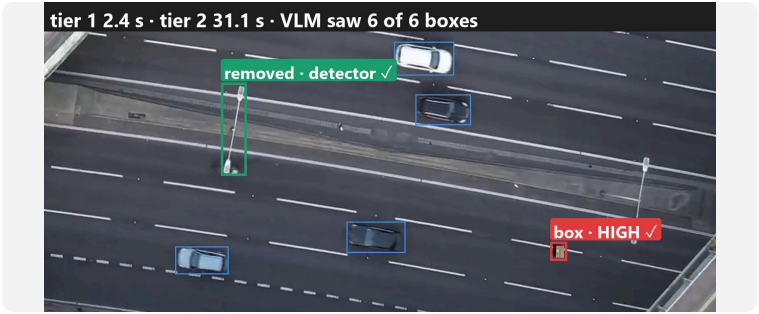
✓ planks alerted · but named 'debris'



X lorry cab → 'barrel' · HIGH · false alarm



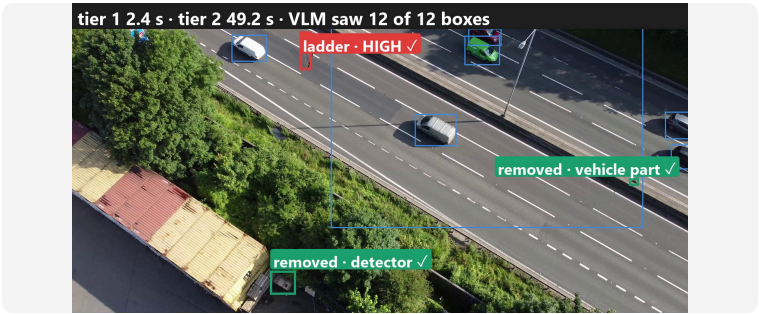
✓ tyre · HIGH · VLM saw all 29 boxes



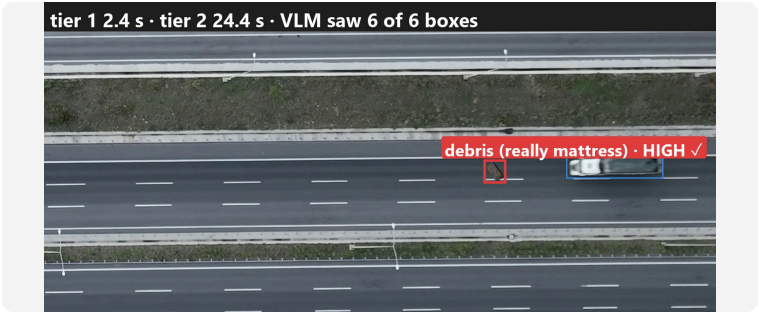
X empty box rated HIGH · rubric: low



X fridge → 'roadside structure' · missed



✓ ladder · HIGH · misnamed boxes left to the gate



X mattress named 'debris' · HIGH · rubric: medium



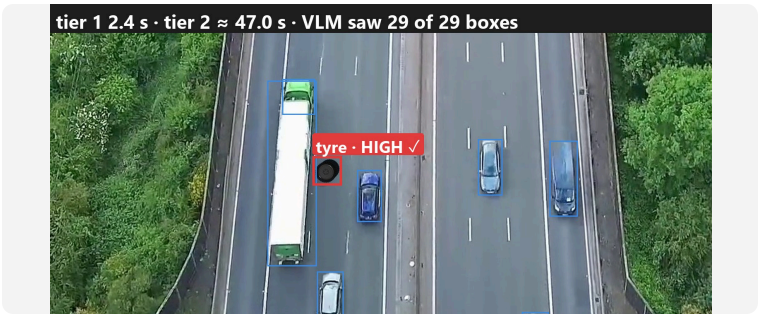
✓ small tyre · HIGH · B, D miss it

B · GPT-5.4

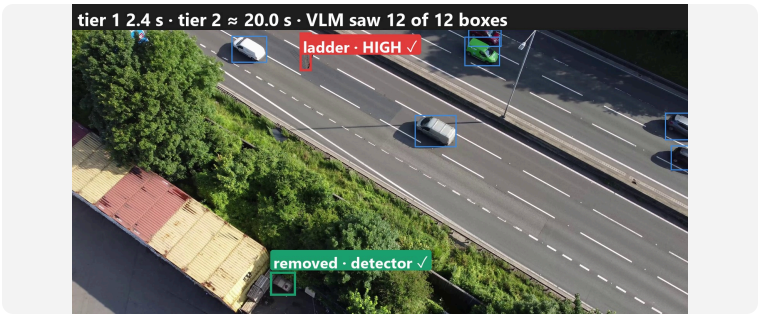
names and rates debris best · no false alarm · but asks the VLM about every box: tier 2 6-49 s
■ high ■ medium ■ low ■ removed by gate ■ missed ■ vehicle · banner: this frame's tier 1 and tier 2 time, and boxes the VLM saw



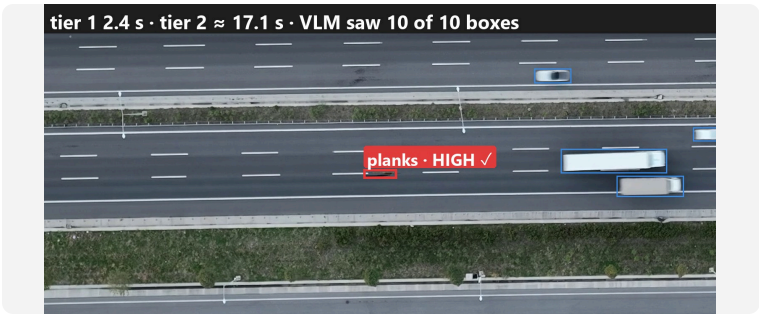
✓ tyre · HIGH



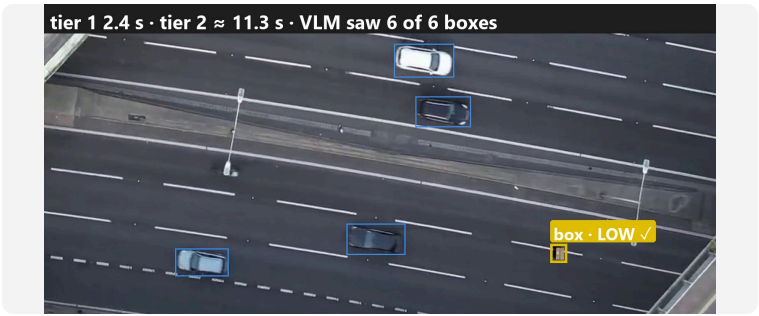
✓ tyre · HIGH · VLM saw all 29 boxes



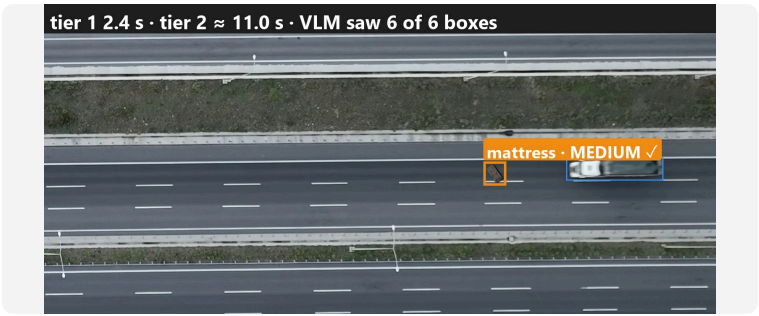
✓ ladder · HIGH · misnamed boxes left to the gate



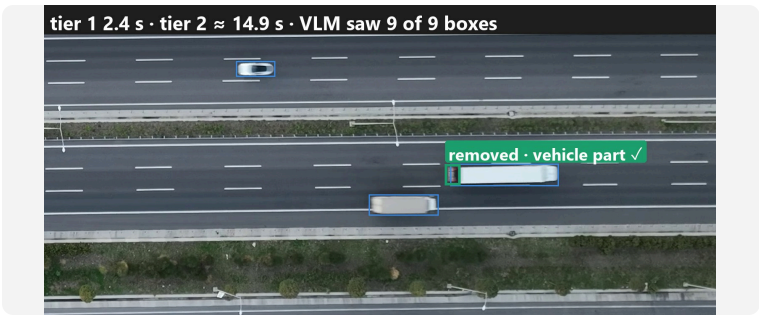
✓ planks · HIGH



✓ box · LOW · as the rubric says



✓ mattress · MEDIUM · as the rubric says



✓ lorry cab removed · GPT: part of the vehicle



✓ fridge · HIGH



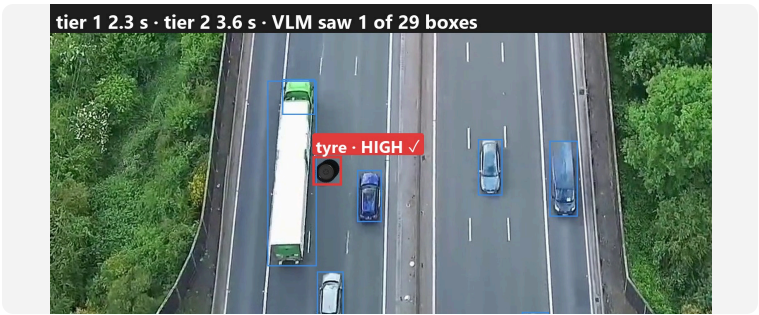
X small tyre → 'roadside structure'

D · detector screens first, GPT-5.4 on flagged boxes

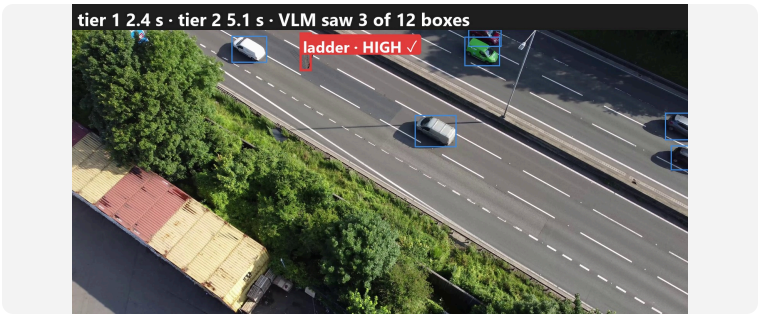
B's answers on every frame · the VLM sees 1-3 boxes instead of all · tier 2 2-7 s
■ high ■ medium ■ low ■ removed by gate ■ missed ■ vehicle · banner: this frame's tier 1 and tier 2 time, and boxes the VLM saw



✓ as B · VLM saw 1 of 5 boxes



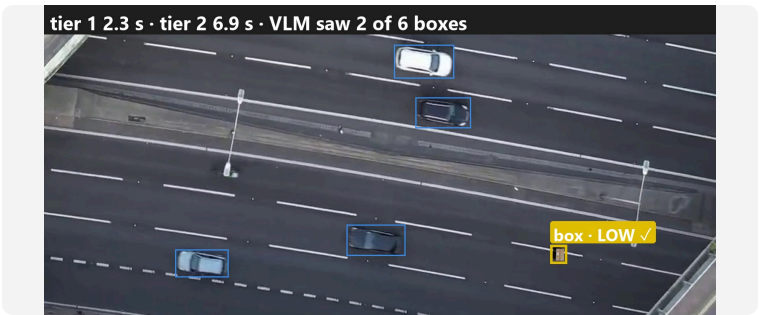
✓ as B · VLM saw 1 of 29 boxes



✓ as B · 3 of 12 · nothing misnamed



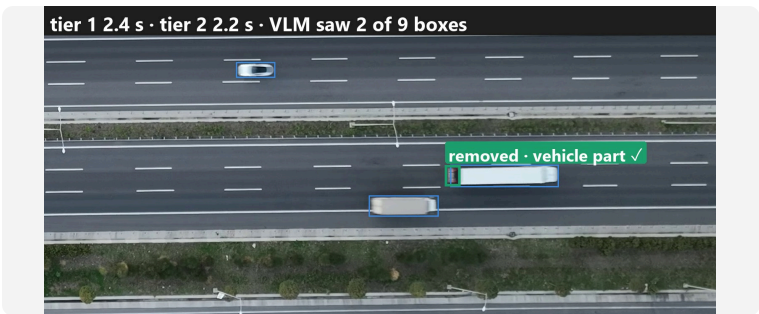
✓ as B · 3 of 10 boxes



✓ as B · 2 of 6 boxes



✓ as B · 1 of 6 boxes



✓ as B · cab removed · 2 of 9 boxes



✓ as B · 1 of 3 boxes



X as B · 1 of 30 boxes

Time, accuracy and GPU

	A · local Qwen	B · GPT-5.4	D · screen + GPT-5.4
Tier 1 · look · every frame, on the drone's GPU · s per frame, 30 test frames, RTX 3050 Ti (4 GB)			
road segmentation object detection (OWLv2), run side by side	2.36	2.36	2.36
re-scoring (+ screen in D)	< 0.01	< 0.01	< 0.01
TIER 1 TOTAL · can the drone keep up?	2.4	2.4	2.4
Tier 2 · assess · A, B: every frame · D: only frames tier 1 flags (27 of 30 here, all hold debris)			
identification (VLM) · bottleneck	49.2	22.0	2.1
context gate	0.09	0.13	0.05
risk assessment (VLM)	4.45	≈ 1.3	1.8
tier 2 total	53.7	≈ 23.4	4.0
VLM calls per frame	16.2	14.9	2.3
END TO END, s per frame	≈ 56	≈ 26	6.3
Accuracy · 30 test frames			
detection: vehicles · debris · precision	95.2% · 90.0% · 84.1% (shared)		
debris named correctly	56%	64%	64%
vehicles recognised	96.0% (VLM)	96.4% (VLM)	98.5% (detector)
debris boxes assessed for risk (of 27) · real debris lost by gate or screen	22 · 0	21 · 0	21 · 0
false alarms rated high	1	0	0
GPU memory · peak			
on the device	2.19 GB tier 1 · 1.97 GB VLM	2.19 GB	2.31 GB
all models loaded at once	4.05 GB > 4 GB card	2.19 GB	2.31 GB
Deployment			
API cost per frame · runs offline	\$0 · yes	\$0.036 · no	\$0.0056 · no